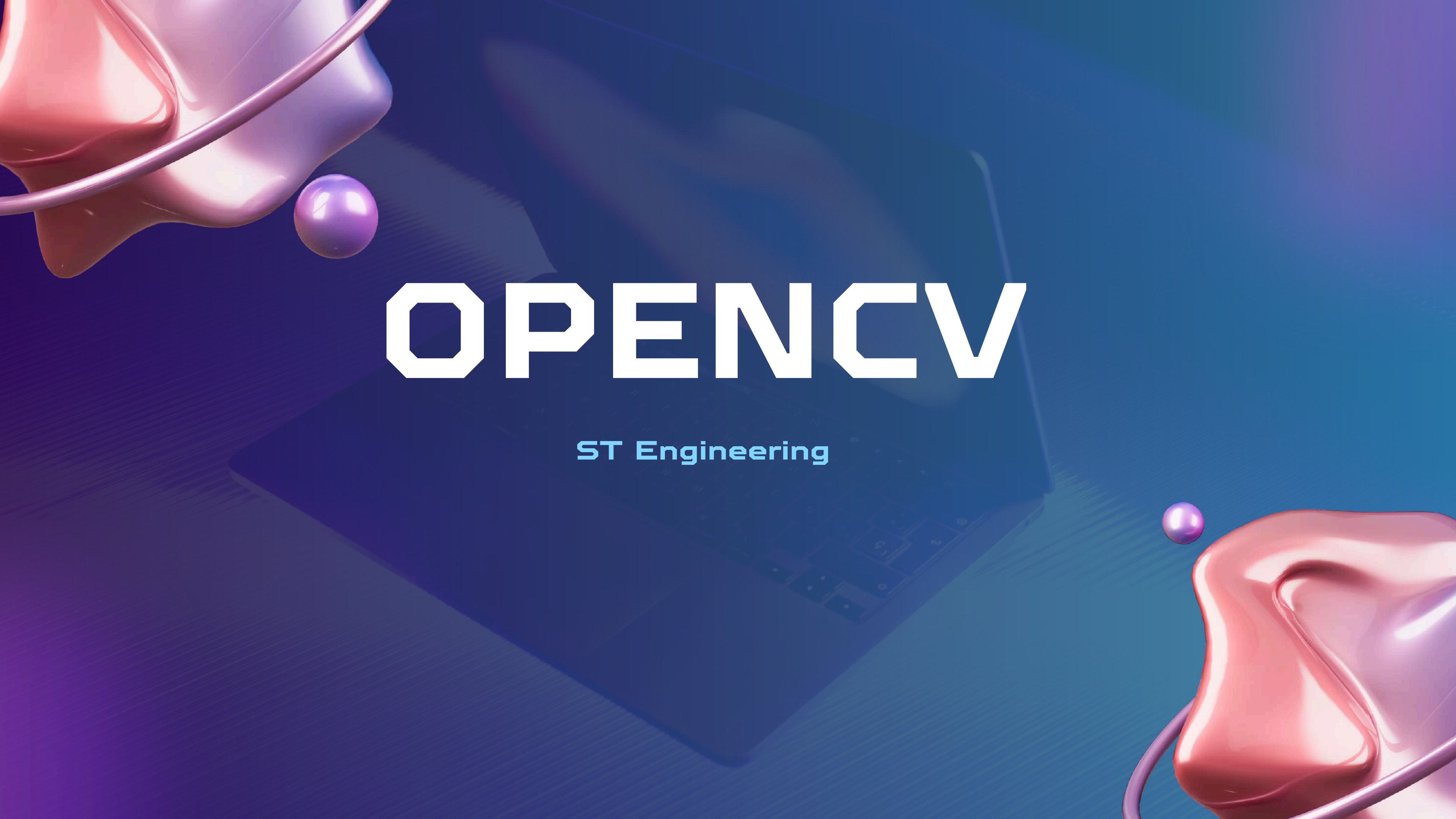


The background is a deep blue gradient. In the center, a laptop is faintly visible, its screen and keyboard area slightly darker. In the top-left and bottom-right corners, there are large, metallic, reddish-pink shapes that look like stylized planets or moons with rings. Small, shiny purple spheres are scattered around these larger shapes.

OPENCV

ST Engineering

DISCLAIMER

The information provided here is for general knowledge and informational purposes only. I am not a qualified expert in this area, and this content should not be construed as professional advice. Always consult with a qualified professional for specific guidance.





IMAGE?

```
[[ 76 112 71]  
 [ 76 112 71]  
 [ 76 112 71]  
 ...  
 [ 76 112 71]  
 [ 76 112 71]  
 [ 76 112 71]]]
```

□

Image Array

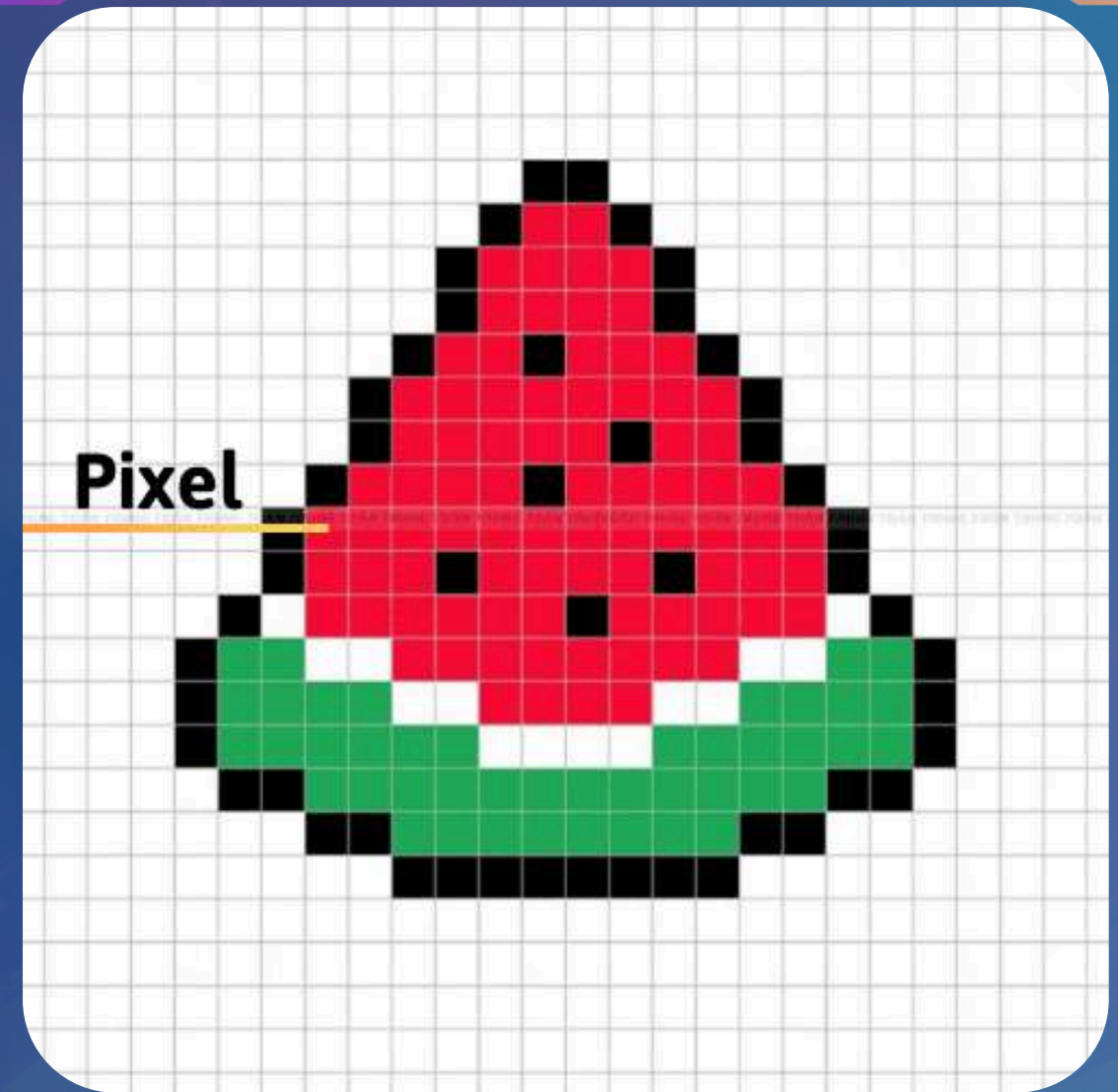


IMAGE SMOOTHING

This is done by convolving an image with a normalized box filter. It simply takes the average of all the pixels under the kernel area and replaces the central element. This is done by the function `cv.blur()` or `cv.boxFilter()`. Check the docs for more details about the kernel. We should specify the width and height of the kernel. A 3x3 normalized box filter would look like the:

$$K = \frac{1}{9} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

IMAGE SMOOTHING

$$\textit{MedianFilter} = \frac{a(\frac{N}{2}) + a(\frac{N}{2} + 1)}{2} \quad (1)$$

IMAGE SMOOTHING

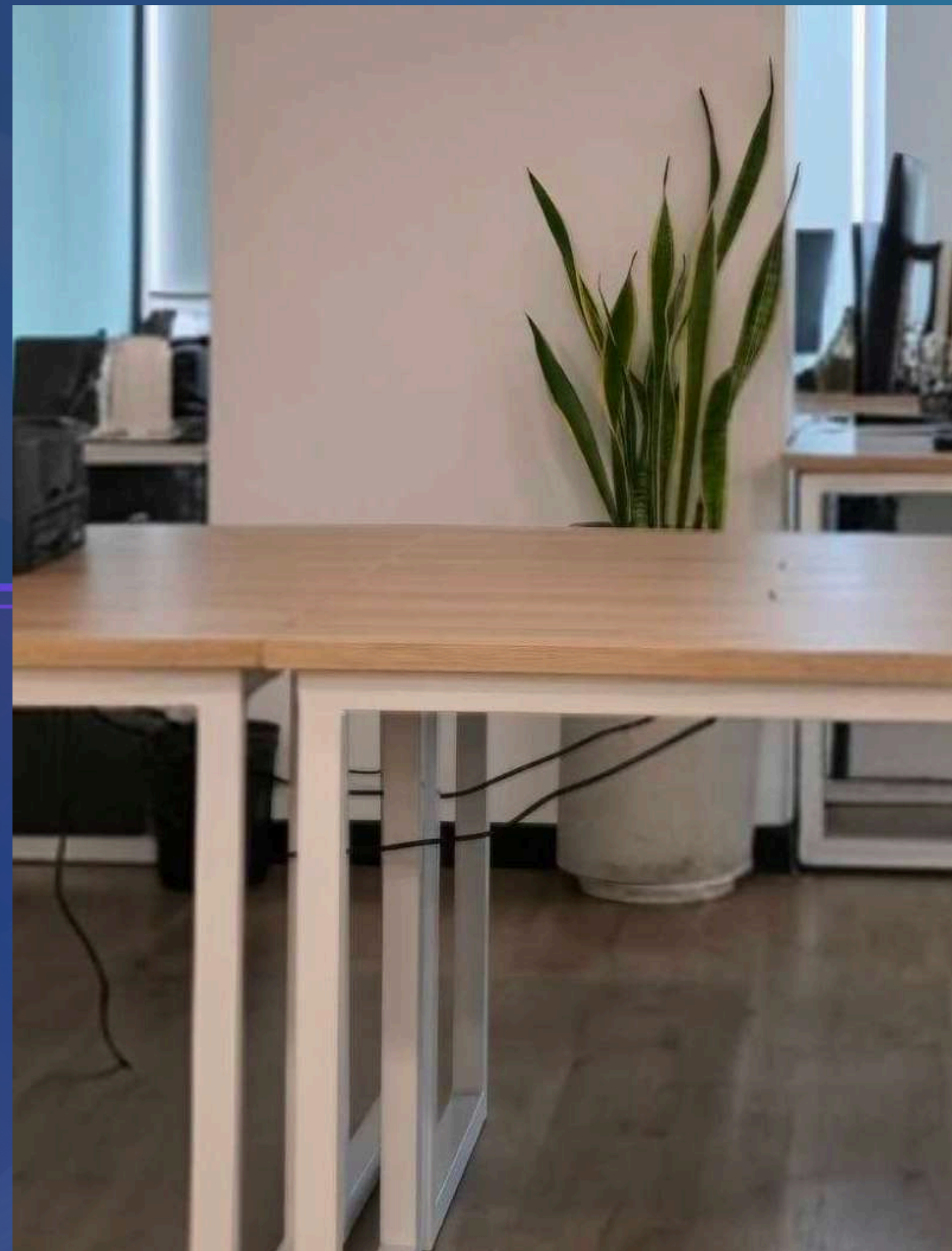
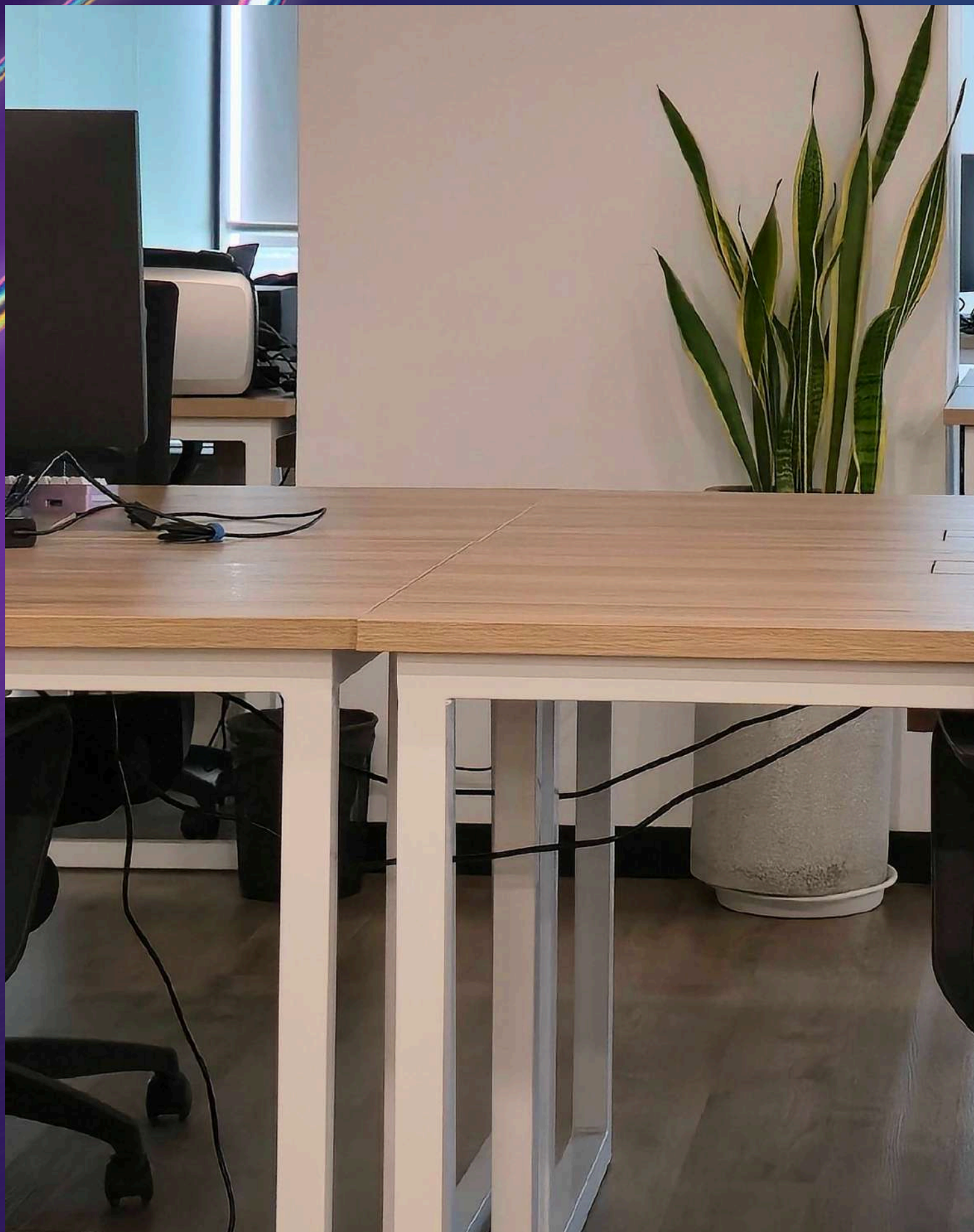
Gaussian Blur

Obtain filter coefficients from sampling 2D Gaussian

$$f(i, j) = \frac{1}{2\pi\sigma^2} e^{-\frac{i^2+j^2}{2\sigma^2}}$$

- Produces weighted sum of neighboring pixels (contribution falls off with distance)
 - Truncate filter beyond certain distance

$$\begin{bmatrix} .075 & .124 & .075 \\ .124 & .204 & .124 \\ .075 & .124 & .075 \end{bmatrix}$$



MORPHOLOGICAL TRANSFORMATIONS

Erosion

We create custom software solutions using cutting-edge technologies for web, mobile, and enterprise needs.

Dilation

We craft and execute digital strategies to optimize online presence, enhance engagement, and drive transformation.



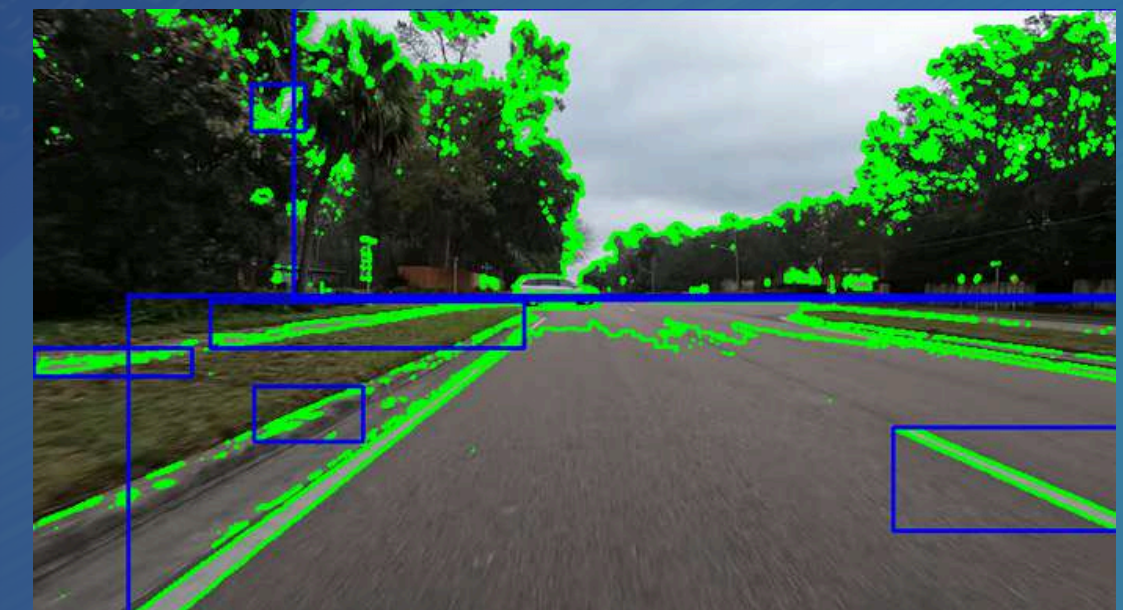
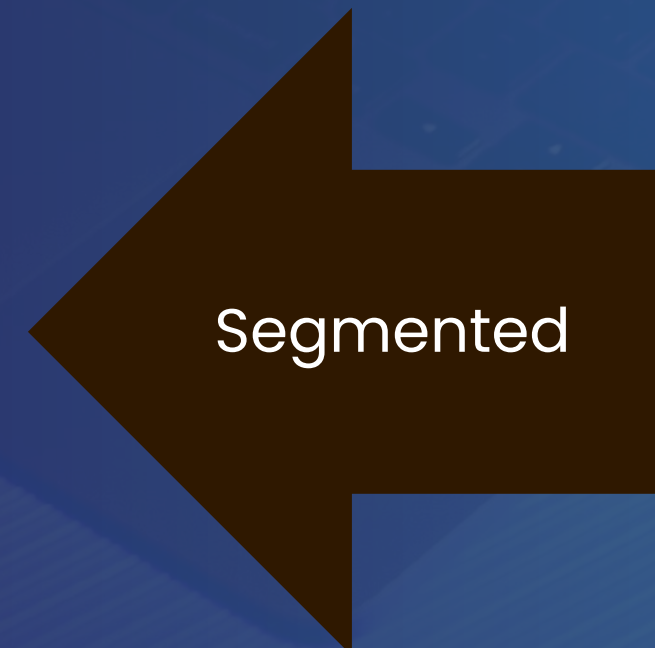
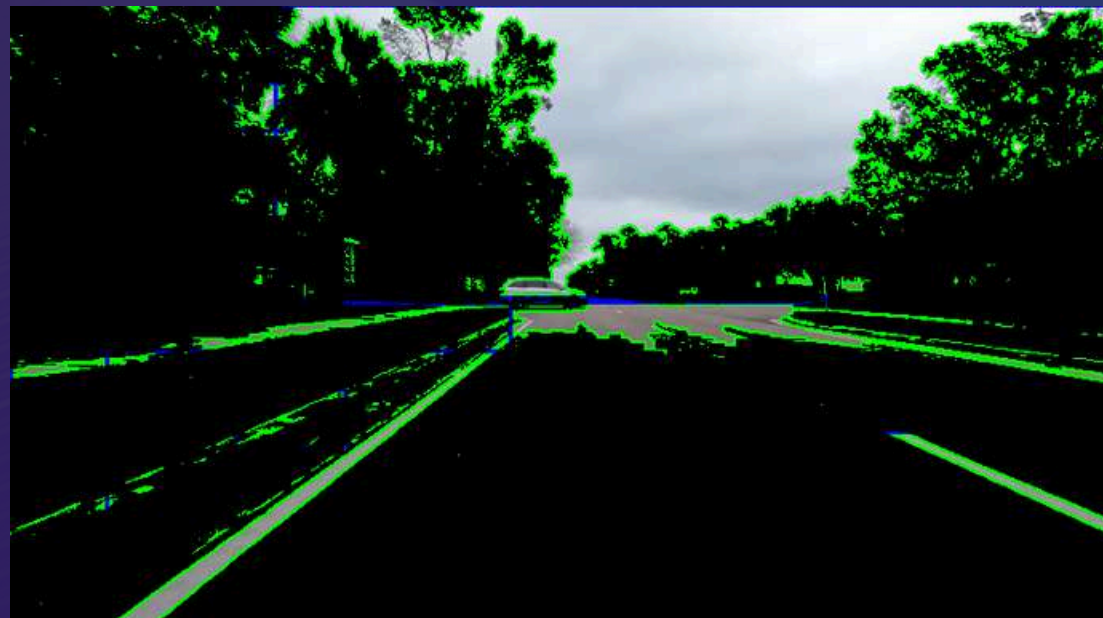
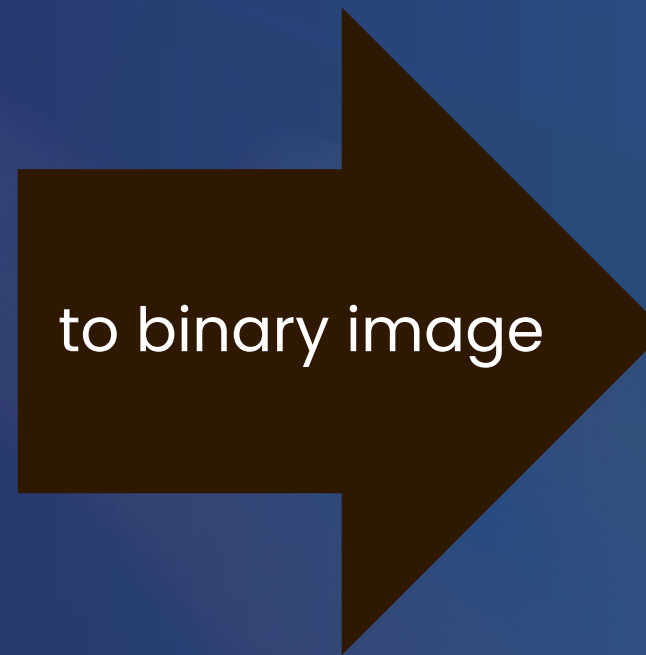
IMAGE SEGMENTATION

Detecting driving lane line using:

- grayscale
- threshold
- binary image
- contours, hierarchy

s of Curvature :463.0
nce to Center -0.429







FEATURE DETECTION



Color



Shape

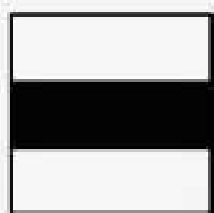
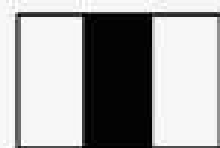


Inside image

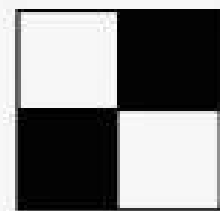
OBJECT DETECTION



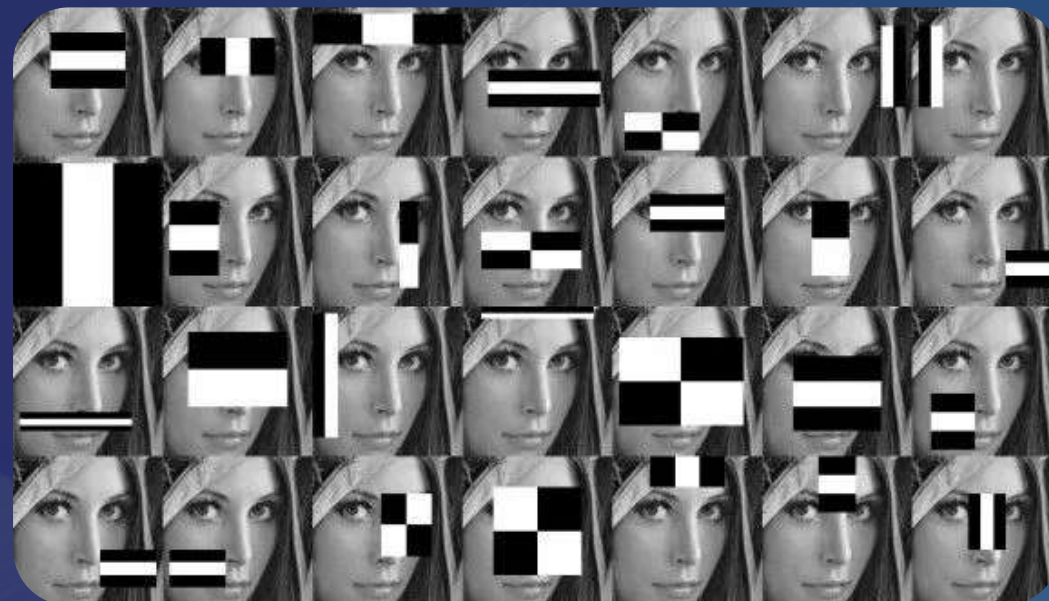
(a) Edge Features



(b) Line Features



(c) Four-rectangle features



- **Haar-like Features:** Using simple rectangular filters to identify differences in pixel intensity (like edges or bars) in an image.
- **Integral Images:** Employing a technique to quickly calculate the sum of pixel values within any rectangular area, significantly speeding up feature evaluation.
- **Adaboost:** Selecting the most relevant features from a large set and combining multiple "weak classifiers" (based on single features) into a strong classifier.
- **Cascade of Classifiers:** Organizing the strong classifiers in a cascade structure. Simpler classifiers are applied first to quickly discard negative regions (no object). Only if a region passes all stages is it classified as containing the object.

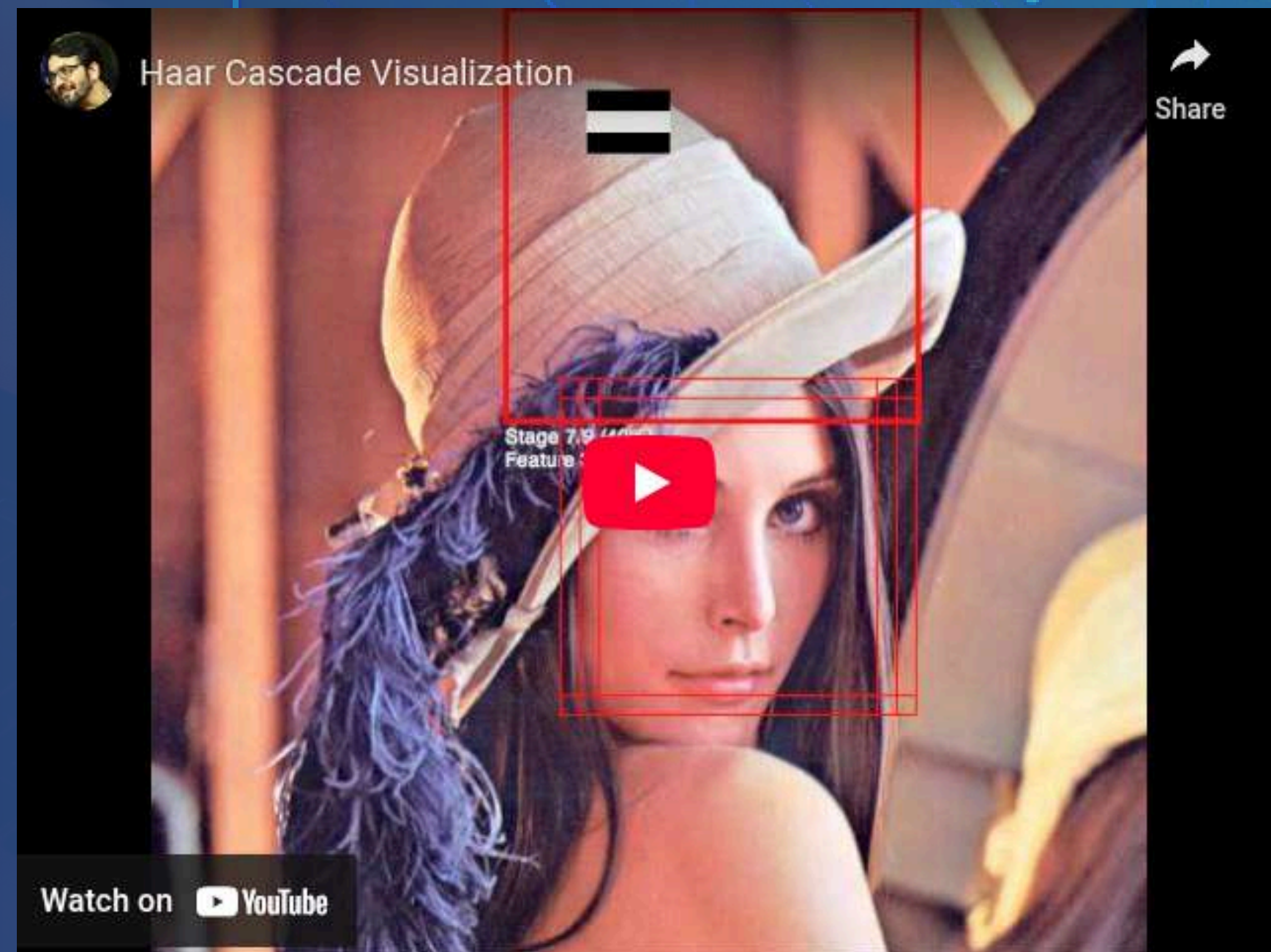


Calculating Haar Features

Creating Integral Images

Adaboost Training

**Implementing Cascading
Classifiers**



Haar Features for Face Detection | Face Detection

Detecting Faces of Different Size

Compute Haar Features at different scales to detect faces of different sizes.

Watch on YouTube

Share

Integral Image | Face Detection

Integral Image

A table that holds the sum of all pixel values to the left and top of a given pixel, inclusive.

98	110	121	125	122	120
89	110	120	116	116	129
87	109	124	111	123	134
88	112	132	108	123	133
87	113	147	108	125	142
95	111	168	122	130	137
96	104	172	130	128	130

Image I

98	208	329	454	576	705
89	658	898	1137	1385	
87	988	1340	1701	2093	
88	392	833	1330	1790	2274
87	488	1043	1687	2255	2864
95	584	1249	2061	2751	3490
96	680	1448	2433	3253	4118

Integral Image II

Watch on YouTube

Share

AdaBoost, Clearly Explained

AdaBoost....

...Clearly Explained!!!

Watch on YouTube

Share

ST ENGINEERING

Q & A

My code doesn't have bugs, it just has unexpected features.



ASSIGNMENT

Detect the lane line in picture

- **Draw the border**
- **print out the center point of the lane line**

https://github.com/fxanhkhoa/AL-ML-Hackathon/blob/main/training/day-3-opencv/images/assignment_lane_line.png





THANK YOU!

FOR JOINING

